SHEEL SHAH

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Computer Science student shipping **backend and agentic AI systems** onto real hardware: a **63%** lower-latency **ONNX** edge-inference pipeline at Infineon, a multi-agent **FastAPI** dispatch backend, and an open-source LLM context tool with **200+ GitHub stars**. Seeking Fall 2026 and Summer 2027 internships.

EDUCATION

University of California, San Diego — Jacobs School of Engineering **Incoming Fall 2026**
B.S. Computer Science, transfer student *Expected June 2028*

De Anza College **Sept 2024 – June 2026**
A.S. Computer Science and Mathematics, Magna Cum Laude | Dean's List, 7 consecutive quarters | Phi Theta Kappa *GPA: 3.91*

Coursework: Data Structures and Algorithms, Object-Oriented Programming in C++, Discrete Mathematics, Linear Algebra, Computer Architecture

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, Swift, Dart, Bash **Databases:** PostgreSQL, PL/pgSQL, Supabase, schema design

Backend: FastAPI, REST APIs, WebSockets, uvicorn, async services, event-driven architecture, Supabase Edge Functions, task orchestration

AI and Agents: large language model (LLM) APIs, multi-agent orchestration, Model Context Protocol (MCP), context engineering, ONNX Runtime, model quantization, edge inference, computer vision, MediaPipe, OpenCV

Infrastructure and Frontend: Git, GitHub Actions (CI/CD), monorepo workflows, pre-commit hooks, Linux, Raspberry Pi, React, Vite, Tailwind, Bun

EXPERIENCE

Infineon Technologies — Software Engineering Intern (in partnership with De Anza College) **Jan 2026 – Mar 2026**

- Shipped a real-time driver drowsiness and distraction detection pipeline sustaining **15+ FPS** at **sub-70 ms** latency, fully on-device with no network
- Cut inference latency **63%** below baseline by converting models to **ONNX** and deploying quantized inference to a **Raspberry Pi** edge board
- Streamed driver-alert telemetry from edge hardware to a **SwiftUI** iOS client via the **PostgreSQL** schema, **PL/pgSQL** functions, and **Supabase**
- Maintained a 163-commit Apache-2.0 **monorepo** spanning firmware, ML, iOS, and database, gated by **GitHub Actions CI** and CODEOWNERS review
- Co-authored the **published research paper** and was one of **5 students** selected from De Anza to present the system at **STEM Zone 2026**

Association for Computing Machinery at De Anza — Club President **June 2025 – June 2026**

- Grew the chapter to **60+ members** and lifted attendance **74%** by recruiting officers and running workshops on **Python**, **Git**, and **agent tooling**
- Led **Halo**, a gesture-controlled photo booth co-built with the Human-Computer Interaction Club, replacing the shutter with **MediaPipe** hand-tracking
- Prepared **4 teams of 5 students** to compete in intercollegiate hackathons through team formation, project scoping, and demo practice sessions

De Anza College — Teaching Assistant, Computer Science and Mathematics **June 2025 – June 2026**

- Supported **90+ students** across **12 weekly office hours**, debugging **C++** and **Python** submissions line by line and reteaching core concepts
- Built a custom **TypeScript** application guiding students through the university transfer journey, from course planning to application deadlines

Foothill–De Anza District — Student Government Finance Intern **June 2025 – June 2026**

- Won board approval of a **$1.4M** student budget representing **18,000 students** and their programs, defending it before the district Board of Trustees

PROJECTS

Accordion — Context Management for AI Agents | Python, TypeScript, Claude API **2nd Place, UC Berkeley AI Hackathon 2026**

- Cut agent token usage **40%+** on **SlopCodeBench** with no measurable accuracy loss by building the Conductor, the layer that folds low-value context
- Gave developers direct control over agent memory by rendering a live **LLM context window** as a grid of foldable, pinnable blocks they can steer mid-run
- Earned **200+ GitHub stars** and 2nd place in The Token Company track; the team is now running formal benchmarks toward a research paper

Clear Dispatch — Multi-Agent 911 Dispatch Support | Python, FastAPI, TypeScript **HackDavis 2026**

- Built the **inter-agent messaging layer** that lets four agents hand work to each other to triage 911 call severity, route the nearest unit, and brief dispatchers
- Eliminated all frontend polling by engineering the **FastAPI** backend and **WebSocket** hub broadcasting every agent state transition to a **React** console
- Gated autonomous dispatch of heavy assets behind **human confirmation**, with surge mode auto-triggered by a sliding 60-second call-rate window

Echo — Offline Emergency Alert Network | Dart, Flutter, Bluetooth Low Energy **Hack4Humanity 2026**

- Relayed emergency alerts device to device across **7 hops in under 3 seconds** with no internet or cellular service, over a **Bluetooth Low Energy mesh**

Assistive Robot for Bedridden Patients | Python, MediaPipe, OpenCV **Winner, MVP Support Prize, HackStorm**

- Located patients in 3D space and read their gestures with no wearables or voice input by mapping **MediaPipe** 2D landmarks to real-world coordinates
- Won the **MVP Support Prize**, awarded for unblocking the computer vision pipelines of **4 competing teams** at the event in addition to my own

LEADERSHIP

Village Lead, De Anza Guided Pathways — Promoted twice from Ambassador; advised **12 students** on course and university transfer planning

Treasurer, Human-Computer Interaction Club and English as a Second Language Club — Managed **$1,000+** in combined club and event funds